

125°C Capable Standalone STT-MRAM for High Reliability Applications

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Spin-transfer torque magnetoresistive random access memory (STT-MRAM) offers promise for use at up to 125°C ambient temperatures in automotive, aerospace, and military applications requiring nonvolatility, high write cycling endurance, and resistance to radiation damage. We report on Everspin's latest xSPI-interface STT-MRAM, which passes AEC-Q100 grade 1 qualification tests for operation from -40°C to 125°C. We examine the read and write window performance to this elevated temperature, the associated qualification tests, and a radiation evaluation against total ionizing dose. In addition, we discuss the future STT-MRAM product roadmap including an enhanced NOR Flash expected to operate with over 10 times greater write cycling endurance, over 10 years data retention at 125°C, and significantly reduced system downtime from over-the-air programming updates in the field.

Index Terms— Automotive electronics, Nonvolatile memory, Product development, STT-MRAM

I. INTRODUCTION

SPIN-TRANSFER torque magnetoresistive random access memory (STT-MRAM) has seen widespread development and commercialization as a next-generation standalone and embedded nonvolatile memory [1, 2]. The perpendicular magnetic tunnel junction (pMTJ) bits can be tuned for write cycling endurance and data retention performance including up to no wear-out errors over at least a 10 year life or no thermally-induced data retention flips over 10 years at 125°C. STT-MRAM is based on the pMTJ bit resistance, and the robustness of pMTJ magnetic properties to radiation makes STT-MRAM more suitable for high radiation environments than charge-based memories like DRAM, SRAM, or Flash. As a result, STT-MRAM is useful in harsh conditions requiring high reliability, such as automotive, aerospace, and military applications.

II. STT-MRAM PRODUCT ROADMAP

The first three columns of Table 1 show standalone STT-MRAM from Everspin [3,4] with an xSPI memory interface for industrial, automotive, and high reliability applications with up to 10 years of unlimited read and write endurance for the EMxxLX and EMxxLX HR versions. The low power EMxxLXB SC for industrial control and embedded storage has up to 16Mb and an operating temperature range of -40°C to 85°C and 10 years of data retention at 85°C. The extended grade

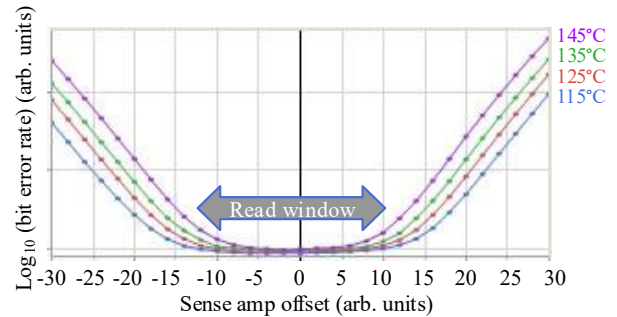


Figure 1: ECC-off bit error rate of the EMxxLX HR high reliability part vs. sense amp offset. The read window around default sense amp offset of 0 decreases with junction temperature from 115°C to 145°C but remains sufficient up to 145°C.

(EMxxLX) has up to 128Mb with operating temperature up to 105°C and data retention to 10 years at 105°C and is targeted for high performance embedded and networking infrastructure. The high reliability EMxxLX HR, in sizes up to 256Mb, targets aerospace, military, and automotive applications in harsh environments and supports AEC-Q100 grade 1 qualification tests for operation up to 125°C with data retention of 10 years at 105°C or 1 year at 125°C.

A next-generation standalone STT-MRAM under development at Everspin shown in the fourth column of Table 1 is targeted as an enhanced NOR Flash with a typical program bandwidth of 90 MB/s and read bandwidth up to 400 MB/s and

Table 1: Current xSPI STT-MRAM from Everspin includes the EMxxLX family. A next generation STT-MRAM is targeted as standalone enhanced NOR Flash.

	EMxxLXB SC (Low power)	EMxxLX	EMxxLXHR (High reliability)	Enhanced NOR
Density	4Mb – 16Mb	32, 64, 128, 256Mb	64, 128, 256Mb	128Mb - 2Gb
Interface	xSPI (x1, x2, and x4)	xSPI (x1, x2, x4, and x8)	xSPI (x1, x2, and x4)	xSPI (x1, x4, and x8)
Max. speed	108MHz STR / 54MHz DTR	200MHz STR / DTR	133MHz STR / 90MHz DTR	Up to 200MHz OSPI (400MB/s)
Endurance	4×10 ¹² write cycles	Unlimited write 10 years	Unlimited write 10 years	> 10× NOR endurance
Data retention	10 years / 85°C	10 years / 105°C	10 year/105°C or 1 year/125°C	> 10 years / 125°C
Operating temp.	-40°C to +85°C	-40°C to +105°C	-40°C to +125°C	-40°C to +125°C
Qual / reliability	Industrial grade	Up to extended grade	Pass AEC-Q100 grade 1 qual tests	Targeting AEC-Q100 grade 1

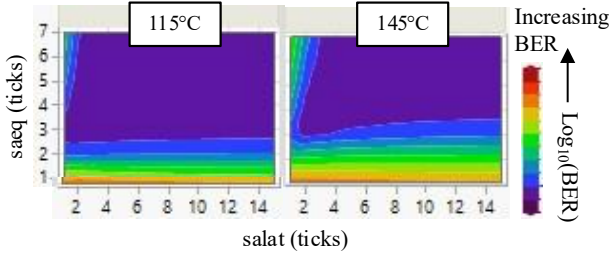


Figure 2. ECC-off bit error rate of the EMxxLX HR high reliability part versus read timing parameters saeq and salat. A wide window to reach the floor bit error rate (BER) exists up to at least a junction temperature of 145°C.

with potential applications for FPGA configuration, AI weights, secure boot, and data and code storage. Initial development samples are currently planned for early 2027. As a native 256Mb and 1Gb die, it will be offered in sizes up to 2Gb with an xSPI interface. Part operation is planned from -40°C to 125°C with over 10 years of data retention at 125°C. Compared to NOR Flash, over 10 times greater write cycling endurance is expected, and the absence of a required erase operation will allow over-the-air firmware programming updates in the field up to 400 times faster.

III. HIGH RELIABILITY xSPI STT-MRAM CHARACTERIZATION AND QUALIFICATION

The read signal in Fig. 1 is critical for operation of the high reliability EMxxLX HR part to 125°C ambient. Decreasing pMTJ tunnel magnetoresistance as temperature increases shrinks the read window. The read circuit operates at a sense amp offset setting of 0. The ECC-off bit error rate increases as the sense amp offset is intentionally varied from 0. The flat bottom width of these curves is the read window, which decreases with increasing temperature, but even at the highest junction temperature of 145°C, sufficient read window exists to reach the bit error rate floor. Similarly, Fig. 2 plots the ECC-off bit error rate vs. read timing parameters saeq and salat. A wide read window exists to a junction temperature of 145°C. The bit-to-bit spin-transfer torque write voltage distributions are plotted in Fig. 3 on a normal quantile scale for both switching directions at junction temperatures of 25°C and 145°C. Each curve is a different die. The write voltage distributions show a well-behaved linear trend to a minimum bit error rate floor.

In order to ensure their high reliability, all EMxxLX HR parts are tested across the full temperature range from -40°C to 125°C ambient as a production screen and are put through an extended burn-in step. During qualification testing for AEC-Q100 grade 1, 80 dies per lot were selected for a high temperature operating life (HTOL) test. The dies were first precycled at the maximum voltage VDD for the part, and then a standard data pattern was written to the array. After exercising the parts for 1000 hours of HTOL at 125°C ambient temperature, the preset data pattern was read back, and functional testing for -40°C, 25°C, and 125°C ambient was confirmed without errors. Early life failure rate (ELFR) stressing at 125°C ambient passed with 800 dies per lot. Finally, high temperature data retention (HTDR) was run with 500 dies per lot and similar precycling to HTOL. A standard memory pattern was written prior to a

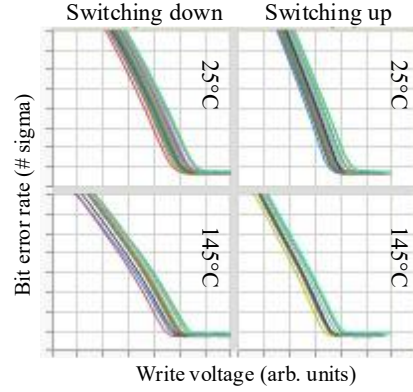


Figure 3. ECC-off bit error rate of the EMxxLX HR high reliability part on a normal quantile scale varies linearly with write voltage for both switching directions up to junction temperature 145°C.

powered-off data retention bake of 135°C for 1000 hours. After the bake, the memory pattern was verified with no errors.

Total ionizing dose (TID) radiation was studied [5] in Everspin's ST-DDR4 1Gb part, which uses the same CMOS platform and a similar pMTJ bit as the EMxxLX. The dies remained fully functional with no ECC-on errors to at least 400 krad(Si) for an active-idle test and 1000 krad(Si) for a static test. For dies exposed to 500 krad(Si), there were no ECC-on errors for write cycling over a 500Mb/10Mb/100kb array space up to $10^6/10^8/10^{10}$ cycles or for 125°C / 312 hour data retention bakes.

IV. CONCLUSION

The EMxxLX HR high reliability STT-MRAM passes AEC-Q100 grade 1 qualification tests for use from -40°C to 125°C. The read and write windows offer margin for operation even above 125°C junction temperature. Qualification testing to ensure operation in this range included HTOL, ELFR, and HTDR according to standard conditions. The STT-MRAM product roadmap also includes an enhanced NOR Flash expected to operate from -40°C to 125°C plus 10 years data retention at 125°C, over 10 times greater write cycling endurance than NOR Flash, and greatly reduced downtime for over-the-air firmware programming updates.

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